

Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Do you know how Java works?



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Explain briefly how different programming languages work.

Can Do Statements:

- I can explain what java is.
- I can describe how java works.
- I can identify and understand key features and concepts related to java.

Ask your partner:

- Why do you think Java is so important in programming?
- Why do you think programmers use Java for programming most of the apps nowadays?





Let's Explore!

Danny and Michelle are studying everything about Java



Instructions: Read the text below.

- Michelle:** Danny, I need your help! I didn't come to class! Could you help me?
- Danny:** Sure! We read everything about Java. You may be confused if you come to class tomorrow without things clear. Let me try to recap everything so you can take notes!
- Michelle:** That'd be great! I heard Laura was so bewildered by it that I'm pretty sure it must be very difficult!
- Danny:** We'll see! Let's start by defining some basic concepts. Do you know what a code is?
- Michelle:** A code is something we use to allude to the statements that we write in a programming language.
- Danny:** Well, you must have some basic concepts clear, so let's move on to Java's definition. Java is a general-purpose computer programming. We consider it as an object-oriented program, which means that we might create programs constructed in Java in terms of objects. Likewise, we see it as a generic program and its platform supports concurrent programming. Now, you might want to know how it works. Java works with a platform-independence since it manages to reorganize the compile-link-execute sequence. Remember that the compiler is the one in charge of turning the statements you wrote in a language into codes. With this implementation, developers succeeded in isolating the dependent issues. Thus, its compiler, well-known as JVM compiler, creates a bytecode which is an object file that works for a virtual machine. Each platform has a Java interpreter that links to a specific code. Compiling this bytecode may help you get Java programs to be faster... Well, after this explanation you can't feel as lost as you were before! You could be confused, but you will get it!!
- Michelle:** Thanks Danny! Of course, I'm not confused now! You could be a great teacher! However... why don't we talk about it one more time?



Let's Practice!

True facts about Java

Instructions: Read the following sentences based on the text. Circle True or False.

1. Java is a general-purpose computer programming and an object-oriented program.
True False
2. They created Java in terms of classes.
True False
3. It is specifically designed to have as few implementation dependencies as possible.
True False
4. Java is also a class oriented program.
True False



Let's Practice!

Don't forget how useful Java is.

Instructions: Select the correct answer.

1. What other language is similar to Java?
 - a. Java is an object-oriented language similar to Spanish
 - b. Java is an object-oriented language similar to C++
 - c. Java is an object-oriented language similar to all languages
 - d. Java isn't a similar to any language
2. Where can Compiled Java run?
 - a. Compiled Java code can run on most computers
 - b. Compiled Java code can't run on any computers
 - c. Compiled Java code can run on most televisions
 - d. Compiled Java code can't run on most hardware
3. Can you convert bytecode to other languages?
 - a. No, bytecode can't also be converted to syntax
 - b. Yes, bytecode can also be converted directly into C++
 - c. No, bytecode can't also be converted to other languages
 - d. Yes, bytecode can also be converted directly into machine language instructions.



Extension Activity

Java virtual machine

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Investigate about Java virtual machine.
- Write down your findings about Java virtual machine.
- Create a poster showing your finding in a creative way.
- Share your findings about Java virtual machine with your teacher and classmates.

Self-Assessment

- | | Yes | No |
|--|-----------------------|-----------------------|
| 1. I investigate about Java virtual machine. | ☐ | ☐ |
| 2. I write information about Java virtual machine. | ☐ | ☐ |
| 3. I create a poster showing my findings. | ☐ | ☐ |
| 4. I show my findings in a creative way. | ☐ | ☐ |
| 5. I talk about Java virtual machine. | ☐ | ☐ |



Target Words

- **allude** (verb): to make reference to something
e.g. A code is something we use to allude to the statements that we write in a programming language.
- **likewise** (adverb): in addition
e.g. Likewise, we see it as a generic program and its platform supports concurrent programming.
- **manage to do something** (noun): to achieve something despite the difficulties
e.g. Java works with a platform-independence since it manages to reorganize the compile-link-execute sequence.
- **recap** (verb): a synonym for summarize
e.g. Let me try to recap everything so you can take notes!
- **bewildered** (adjective): to be confused or disoriented about something
e.g. I heard Laura was so bewildered by it that I'm pretty sure it must be very difficult!
- **isolate** (verb): to set apart or separate
e.g. With this implementation, developers succeeded in isolating the dependent issues.



Learning Points

Modal verbs for probability: might, could, may must and can't

Use the modal verbs to express certainty, probability and possibility about an event. In this case, you are going to learn 5 modal verbs to express probability.

Depending on the degree of certainty we have regarding an event, we can use **might**, **could**, **may**, **must** and **can't** to make a guess about how probable it is to happen.

Remember the following structure when you are writing a sentence with a modal verb:

Subject + **modal verb** + **verb in infinitive form** + complement.

Might, **may** and **could** express the same as **maybe**.

Example:

Sara **might be** at Laura's house. *(Maybe she is there.)*

Laura **may want** to know more about Java. *(Maybe she is interested in knowing more information about it.)*

Juan **could study** with you today. *(Maybe he has the time.)*

Use **must** when you are sure about the event you are guessing about.

Example:

Laura **must be** at Sara's home by now. *She left an hour ago!*

(The person is sure about it and the time is a good reason to support it.)

Use **can't** when you want to express that something is not possible.

Example:

Laura **can't be** at home right now. *She said she had to be there at 3 o'clock!*

(The person knows it isn't possible that Laura is still at home.)

Examples

- *You may be confused if you come to class tomorrow without things clear.*
- *I'm pretty sure it must be very difficult.*





Learning Points

Common content words and expressions used in Java

To understand what Java is and how it works you should have in mind some technical concepts. Here you have some of them:

Class-based programming: It is also known as object-oriented programming (OOP). An OOP is a software design method based on the concept of objects. These objects have certain characteristics: the state, which refers to what the objects have; behavior, which stands for what they do; and identity, which is that special characteristic that makes objects unique.

Concurrent programming: It is a form of modular programming which doesn't require multiprocessors. As a server, it executes several streams of operations concurrently in overlapping periods of time to achieve a common goal.

Structured programming: It is also a form of modular programming which aims to execute logical structures so that the program could be more efficient and easy to modify. You can use it in any language to avoid pitfalls of unstructured ones.

Generic programming: To develop efficient software libraries, they use this programming paradigm. It makes programs more general so that they can be more adaptable. As its algorithms are generic, they can be combined in several ways to produce a great variety of useful software.